

Deixis, entropy, and the architecture of grammar

ABSTRACT: We propose a principled account of deixis as a structural property of grammar that is independently required for the constitution of propositional meaning and whose systematic encoding becomes indispensable under the constraints imposed by externalization. Building on an Un-Cartesian model, we assume that grammar operates over intensional representations derived from world-indexed states. Once such representations are externalized into a temporally unfolding signal, encoding under constrained resources does not preserve all aspects of the original representation, yielding irreversibility in the sense of Shannon's information theory. As a consequence, linguistic signals underdetermine their interpretation, giving rise to a recovery problem. We argue that this problem specifically concerns the anchoring of content to a context of use, and that, since this is precisely the dimension lost in encoding, recovery requires reintroducing these parameters, which grammar encodes through deictic features such as person, time, and location. On this view, deixis is not a peripheral class of expressions but a fundamental component of grammar that both enables propositional content and ensures its recoverability under externalization.

1. Introduction: The problem of context-dependence

It is a familiar and widely acknowledged fact that linguistic interpretation is systematically dependent on context. The meaning of an utterance cannot, in general, be determined solely from the lexical items it contains and the way they are syntactically combined. Rather, interpretation requires reference to features of the context of use, including the identity of the speaker, the time of utterance, and the spatial and perspectival coordinates relative to which the utterance is evaluated. This phenomenon has been extensively documented in the linguistic and philosophical literature on deixis and indexicality (e.g. Bühler 1934/1982; Lyons 1977; Levinson 1983, 2004; Kaplan 1989;¹ Perry 1993). In standard approaches, however, context-dependence is typically treated as a property of a restricted class of expressions. On this view, deixis is associated with so-called indexicals, i.e. pronouns, demonstratives, and certain adverbials, whose interpretation varies systematically with context. Influential accounts in this tradition, such as those of Kaplan (1989) and Perry (1993), sharply distinguish between expressions whose content is fixed independently of context and those whose interpretation requires reference to contextual parameters.

Yet this restricted view of deixis sits uneasily with a range of well-known facts. Even sentences that do not contain any overt indexical expressions systematically fail to determine a unique interpretation in the absence of contextual anchoring. A simple example such as *John arrived yesterday* leaves open which individual named John is intended, what temporal interval counts as *yesterday*, and from which perspective the event is described. Similar considerations apply to proper names more generally, as well as to tense, aspect, and a wide

¹ Throughout this work, Kaplan (1989) is a shortcut for Kaplan (1989a,b).

range of constructions whose interpretation depends on anchoring to a speaker-relative context (Sigurðsson 2004, 2017). In this sense, context-dependence is not confined to a marginal class of expressions, but appears to be a pervasive property of linguistic interpretation (Travis 1997; Recanati 2004, 2010). This raises a more fundamental question: why is grammar inherently context-dependent? More specifically, why does the human language faculty systematically require anchoring to parameters such as person, time, and location in order to yield determinate interpretations? While existing accounts describe how context enters into interpretation, they do not, in general, explain why linguistic systems should be organised this way. For instance, Kaplan's (1989) account formally characterizes the context-dependence of indexicals but treats it as a lexical property of a restricted class of expressions. Similarly, Davidson (2005) emphasizes the role of context in fixing truth conditions, but does not derive why linguistic systems should systematically require such anchoring. What remains unexplained in these accounts is why context-dependence is pervasive and structurally encoded across grammar.

In this paper, we propose a principled answer to this question. Building on the view that grammar is fundamentally a system for establishing reference (Hinzen & Sheehan 2015), we argue that context-dependence is not an accidental feature of language use, but a necessary consequence of the way linguistic representations are encoded and transmitted. Our central claim is that the necessity and systematic encoding of deixis can be derived from general constraints on representation and externalization, the latter as characterised in information theory (Shannon 1948, 1949; see also Fortuny & Corominas-Murtra 2013 for applications to language). At the same time, we suggest that deixis plays a dual role in the architecture of language. On the one hand, it provides the anchoring required for predication to yield truth-evaluable propositions, and is therefore a necessary component of the cognitive architecture of language. On the other hand, as we will show, it becomes indispensable under the conditions imposed by linguistic communication, where reference must be recovered from an underspecified signal. In particular, we argue that the process by which world-indexed representations are transformed into linguistic signals does not preserve all aspects of the original representation, and that successful interpretation therefore requires access to information that restores the link between linguistic content and the world. We propose that this recovery mechanism is grammatically implemented as deixis. Crucially, we argue that this is not an arbitrary design choice: since what is lost in the course of encoding concerns the anchoring of content to a speaker-relative context, any mechanism capable of restoring reference must reintroduce precisely these parameters. More generally, we suggest that the pervasive context-dependence of linguistic meaning follows from third-factor principles governing efficient representation and transmission, rather than from language-specific adaptations (Chomsky 2004a,b). In this sense, the present account is compatible with the view that the faculty of language is primarily optimized for the conceptual-intentional interface, with externalization playing a secondary role (Berwick & Chomsky 2016), although we will argue that the latter nevertheless imposes non-trivial constraints on the form that grammatical systems take.

In order to sustain this argument, the paper is structured as follows. Section 2 outlines a view of grammar as a system of reference and introduces the operations of deindexation and reindexation. Section 3 presents the relevant information-theoretic concepts, focusing on compression, entropy, and irreversibility. Section 4 then develops the core argument linking information loss to the necessity of deictic encoding. Section 5 discusses the consequences of this proposal and its implications for the nature of grammar. Section 6 concludes.

2. Grammar as a system of reference

A standard assumption in formal semantics, inherited from Frege, is that the meaning of a complex expression is determined by the meanings of its parts and the way they are syntactically combined (Montague 1970; Partee 1984). On this view, compositionality provides the basic architecture for explaining how linguistic expressions come to have the meanings they do. Once lexical meanings are fixed and the rules of combination are specified, the interpretation of larger structures follows. However, this view faces a well-known difficulty: it presupposes that the basic units of interpretation already come equipped with determinate semantic values. As noted in Davidson (2005), compositionality does not in itself explain how expressions come to be associated with truth conditions; rather, it presupposes that such associations are already in place. In this sense, compositionality is a method for recursively computing interpretations, but not a theory of how those interpretations are grounded.

This limitation becomes particularly clear when we consider the pervasive referential underdetermination in linguistic expressions. Even relatively simple sentences fail to determine a unique interpretation in the absence of contextual anchoring. A sentence such as *John arrived yesterday* leaves open which individual named John is intended, what temporal interval counts as *yesterday*, and how the event is situated relative to a speaker. Crucially, this indeterminacy is not resolved by adding further lexical material or syntactic structure: it persists unless the expression is interpreted relative to a context. The conclusion must then be that lexical meaning and compositional structure are not sufficient to yield determinate interpretations. Something beyond the meanings of parts and their mode of combination is required, namely a mechanism that anchors linguistic content to the world. This suggests that grammar cannot be understood solely as a system for combining concepts, but must also be a system for establishing reference.

A framework that takes this conclusion seriously is the so-called Un-Cartesian model of grammar (Hinzen & Sheehan 2015; Hinzen et al. 2024). On this view, grammar is not primarily a combinatorial device for assembling pre-existing concepts into structured thoughts, but a system that creates the very conditions under which reference to the world becomes possible. Rather than operating on independently given representations, grammar provides the architecture through which such representations acquire referential content. Within this approach, grammatical structure is organised in terms of phases (Chomsky 2001, 2008), which can be understood as domains of interpretation (Hinzen 2012). Crucially, phases are not merely computational units, but loci where reference is established. Each

phase exhibits an internal organisation that distinguishes between a descriptive core and a referential periphery. This distinction can be captured in terms of an edge-interior structure. The interior of the phase contains descriptive content, i.e. lexical material that classifies or characterises entities, events, or states of affairs. The edge, by contrast, is associated with referential anchoring: it provides the means by which the descriptive content is related to the world. In this sense, grammatical structure systematically separates what is described from how it is anchored. Importantly, this architecture is not confined to a single domain. Parallel structures can be identified across the nominal, verbal, and clausal domains, suggesting a unified system of reference that operates across different levels of grammatical organisation (Sheehan & Hinzen 2011; Arsenijević & Hinzen 2012). In each case, the same basic pattern emerges: lexical content provides a description, while functional structure provides the means to anchor that description to a referential context.

If grammar is a system for establishing reference, then it must mediate between two distinct types of representation: on the one hand, representations that are directly tied to the world, and on the other, representations that are suitable for internal computation. This mediation can be understood in terms of two complementary operations: *deindexation* and *reindexation* (Hinzen & Sheehan 2015: ch. 2). *Deindexation* is the process by which representations are decoupled from their immediate dependence on the world. Perceptual and conceptual representations are initially indexed to particular stimuli or states of affairs: they are, in this sense, extensional. In order to be manipulated by the grammatical system, however, they must be transformed into intensional objects, i.e. abstract representations that can be combined and recombined independently of the presence of their referents. This idea has been developed in various forms in the literature (e.g. Boeckx 2010), and captures the intuition that grammar operates over representations that are no longer directly tied to the world. In contrast, *reindexation* is the converse operation. Once intensional objects have been constructed and combined, they must be reconnected to the world in order to yield interpretable expressions. This is achieved by anchoring them to contextual parameters such as speaker, time, and location, thereby restoring a form of referential linkage (Hinzen & Sheehan 2015). Reindexation thus provides the bridge between grammatical structure and the world it is used to describe.

These two operations introduce a fundamental tension at the heart of the grammatical system. On the one hand, computation requires abstraction: representations must be freed from their dependence on particular contexts in order to be manipulated by general rules. On the other hand, reference requires anchoring: in order to yield determinate interpretations, those representations must ultimately be tied back to a specific context of use. Grammar must therefore reconcile these competing demands, simultaneously enabling abstraction and ensuring anchoring.

In the next section, we argue that this tension is not merely conceptual, but has concrete consequences for the design of the system. In particular, we show that the process of *deindexation*, understood as a form of compression, necessarily introduces information loss, and that this, in turn, requires the systematic encoding of recovery mechanisms. We propose that *deixis* is precisely the grammatical realisation of such mechanisms.

3. Information-theoretic constraints on linguistic encoding

The previous section argued that grammar cannot be understood solely as a system for combining meanings, but must also be a system for establishing reference. In particular, we introduced the distinction between deindexation and reindexation, which gives rise to a fundamental tension: grammatical computation requires abstraction from context, while interpretation requires anchoring to it. This tension is not merely conceptual, but reflects general constraints on the encoding of information in finite systems. More specifically, the process of deindexation can be understood as a form of compression, and this necessarily entails information loss. As a consequence, linguistic systems face a recovery problem: the signal underdetermines the intended interpretation. This problem will provide the basis for our account of deixis in Section 4.

Any system that transmits information from one point to another must encode that information into a signal to solve what Shannon called the fundamental problem of communication, namely “*reproducing at one point either exactly or approximately a message selected at other point*” (Shannon 1949: 32).² In the case of human language, the relevant signals are sequences of phonological units realised in time. These signals are necessarily limited in their expressive capacity: they are linear, finite, and constrained by articulatory and perceptual systems.³ By contrast, the representations that language is used to convey, whether characterised as conceptual structures, perceptual states, or truth-evaluable propositions, are comparatively rich and high-dimensional. The mapping from such representations to linguistic signals therefore cannot be a simple one-to-one correspondence. Instead, it must involve a process of compression, whereby a rich representational structure is encoded into a more economical form. This is a general property of communication systems, and it is reflected in well-known features of language such as Zipfian distributions, where more frequent elements tend to be shorter and less marked (Zipf 1965; Ferrer-Cancho & Solé 2003). From this perspective, linguistic encoding can be seen as a process that trades representational detail for efficiency.

Within Shannon’s framework, a first aspect of this problem concerns *source coding*, i.e. the efficient encoding of messages by reducing redundancy in the signal (Shannon 1949: 32). In the limit, optimal source coding aims to represent messages using the minimal amount of information compatible with their probability distribution. Crucially, this process of

² While Shannon’s framework is formulated independently of semantic interpretation, we adopt it here in a reinterpretive sense: the uncertainty it characterizes is taken to correspond, in the linguistic domain, to the underdetermination of interpretation given a signal.

³ We do not assume any privileged channel for the externalization of language. The mapping from internal linguistic representations to external signals can be implemented across different sensorimotor modalities, including the oral-auditory channel characteristic of speech and the manual-visual channel one of sign languages. These modalities differ in their physical realization, but not in the underlying computational system they externalize. More generally, externalization should be understood as a modality-independent interface between the internal linguistic system and the sensorimotor systems. In this sense, what is relevant is not the specific channel, but the fact that linguistic structure must be encoded into a temporally (or spatially) structured signal that can be transmitted and decoded.

compression is not optional. Given the constraints on the physical realisation of linguistic signals and the need for efficient transmission, any viable system must reduce the informational complexity of the representations it encodes. In the terms introduced in the previous section, this means that deindexation, the transformation of world-indexed representations into intensional objects suitable for grammatical computation, can be understood as a compression step. Importantly, when the space of possible signals is smaller than the space of possible inputs, compression entails information loss: distinct inputs may be mapped onto the same output. Formally, this means that the mapping from messages to signals is non-bijective, so that the original input cannot, in general, be uniquely recovered from the output alone. This property can be characterised in terms of *entropy*. In Shannon's Information Theory, the entropy H of a discrete random variable X (formally, $H(X)$) measures the statistical uncertainty associated with its possible values. More precisely, it quantifies the expected uncertainty involved in predicting the outcome of X , and is therefore directly related to information: the greater the uncertainty, the more information is gained upon observing the outcome. Entropy is always non-negative, and provides a measure of the informational content of a source independently of the particular encoding used. As will become relevant below, conditional entropy can be used to quantify the residual uncertainty about the input given the output, thereby capturing the extent to which a signal underdetermines the message.

Importantly, if the conditional entropy of the input given the output is greater than zero, then there is residual uncertainty about the input once the output is known. In other words, the signal underdetermines the message. This is not a contingent feature of particular systems, but a general consequence of constrained encoding: unless the signal space is at least as large as the space of possible messages, some degree of information loss is unavoidable. Formally, and for the purposes of this article, let X_Ω be a random variable taking values in a set of truth-valued (i.e. indexed) propositions $\Omega = \{p_1, \dots, p_n\}$, with probability measure q , where $q(p_k)$ is the probability that $p_k \in \Omega$ is the input to a given computation C . Let X_S be a random variable taking values in a set of signs $S = \{s_1, \dots, s_m\}$, with probability distribution r , where $r(s_i)$ is the probability that symbol s_i is the output of C . We can then define $H(X_\Omega | X_S)$, as the conditional entropy that quantifies the remaining uncertainty about which proposition $p_k \in \Omega$ was the input of C , given that a particular $s_i \in S$ is its observed output. This quantity thus measures the extent to which the signal fails to uniquely determine the message. In information-theoretic terms, such residual uncertainty reflects the limitations of the encoding and transmission process, and provides a principled measure of how much additional structure or redundancy may be required, through mechanisms analogous to *channel coding* (Shannon 1949: 35), to ensure reliable recovery of the intended message.

The immediate consequence of this is irreversibility (Landauer 1961; Bennet & Landauer 1985). Once a message is encoded into a signal under a non-bijective mapping, it cannot, in general, be reconstructed without ambiguity. The decoding process cannot, in general, invert the encoding function, because multiple inputs may correspond to the same output. As a result, interpretation requires access to information beyond the signal itself. Applied to language, this means that linguistic expressions, as physical signals, cannot fully determine

the representations they encode (Fortuny & Corominas-Murtra 2011, 2013). Even if the grammar provides a systematic mapping from underlying representations to surface forms, the resulting signal will not, by itself, uniquely fix the intended interpretation. This is indeed what we observe empirically: linguistic expressions systematically underdetermine their interpretation, and are hence systematically compatible with multiple interpretations, unless supplemented by contextual information.

This irreversibility gives rise to a recovery problem: the signal underdetermines the message and is compatible with multiple possible interpretations. In order to identify a determinate interpretation, the system must provide access to additional information that reduces the uncertainty introduced by encoding. Crucially, this recovery cannot be left entirely to general inference or pragmatic reasoning (Davidson 2005), but requires that at least part of the relevant information be systematically encoded. Otherwise, interpretation would depend on unconstrained inferential processes, undermining the stability and efficiency required for systematic linguistic communication.

The next question is what kind of information is required for recovery. Since the encoding process does not preserve all aspects of the original representation, the missing information must correspond to those dimensions along which the signal underdetermines interpretation. As argued above, the critical dimension concerns the anchoring of content to the world. Linguistic expressions do not, in general, specify which individual is being referred to, when an event takes place, or how the described situation is situated relative to a speaker. These parameters are essential for fixing reference, yet they are not fully encoded in the signal. The recovery problem is therefore, at its core, a problem of anchoring. In order to reconstruct the intended interpretation, the hearer must recover the contextual parameters that situate the content of the expression relative to a context of use, including the identity of the speaker and addressee, the temporal location of the utterance, and the relevant spatial and perspectival coordinates. Critically, this information is not recoverable from the signal alone. It must be supplied from outside the signal, but integrated into the interpretive process in a systematic way. This raises the question of how such information is made available to, and interfaced with, the grammatical system.

If recovery requires access to contextual parameters, and if interpretation is guided by grammatical structure, then grammar itself must play a role in encoding or regulating these parameters. The system cannot rely on external context in an unconstrained way, but must incorporate structured means of interfacing with it. More generally, the design of grammar must satisfy not only the requirements of computation, but also those of recoverability: it must be organized so that information not preserved by encoding can be reconstructed in a reliable and efficient manner.

Importantly for present purposes, communicative situations make the consequences of deindexation and reindexation particularly salient. In even the simplest case of an utterance, a speaker aiming to convey a truth-evaluable proposition must encode it into a temporally linear signal that underdetermines its interpretation. As argued above, this underdetermination arises because the encoding process does not preserve the anchoring of content to a speaker-relative context. Successful interpretation therefore requires that this

anchoring be restored. The question is how the system provides access to the parameters required for such re-anchoring.

We argue that these parameters are encoded through deixis. Since what is lost in encoding concerns the anchoring of content to a speaker, a time, and a perspective, any recovery mechanism must make precisely these dimensions available. Deictic features, traditionally associated with indexical expressions, constitute a general mechanism by which grammar encodes this information. On this view, deixis is not a peripheral or optional component of language, but a structural requirement that follows from the irreversibility introduced by encoding under constrained signal resources.

4. From deindexation to deixis

Deixis, we argue, is not a peripheral or optional feature of grammar, but a necessary consequence of the way linguistic systems encode and recover information under the conditions imposed by externalization. If linguistic representations are externalized for communication and thereby mapped onto an underdetermining signal, then grammar must incorporate mechanisms for restoring the anchoring required for interpretation. In this section, we develop this claim step by step from independently motivated assumptions about grammatical computation and the information-theoretic constraints discussed in the previous section.

Within the biolinguistic tradition, the Strong Minimalist Thesis (SMT) has framed the faculty of language as an optimal solution shaped by interface conditions and third-factor principles of computational efficiency (Chomsky 2004a,b). This perspective rightly emphasizes that linguistic systems are subject to general constraints of design and computation. But it leaves open a more fundamental question: not how language is optimally organized, but how a grammatical system capable of reference arises at all (Hinzen & Sheehan 2015). We argue that third-factor principles constrain the space of possible grammars, but do not by themselves explain the emergence of grammar as a system of reference. That emergence, we suggest, reflects a deeper reorganization of cognition, in which deictic anchoring and propositional structure arise together as properties of a newly integrated system.

Importantly, the information-theoretic considerations invoked here do not model the internal structure of the faculty of language itself, but the constraints imposed on its externalization. Crucially, these constraints are not merely peripheral to interpretation. Once linguistic representations are externalized into a signal, they must satisfy conditions of efficient encoding and recoverability. These conditions do not generate grammatical structure, but determine what information must be made available for interpretation under externalization, and thereby constrain what grammar must encode. These constraints therefore operate at the level of what must be made available for interpretation under externalization, rather than determining the internal generative architecture of grammar itself. In particular, they force the explicit encoding of anchoring parameters, giving rise to the systematic role of deixis.

While the present account assigns a central role to communication in shaping the conditions under which linguistic systems operate, it does not reduce the origin of grammar to communicative interaction (as in Tomasello 2008 or Kirby et al. 2008). Rather, communicative pressures define a problem space, namely the recoverability of reference under conditions of externalization, to which grammar, as a system of reference, provides a structural solution. On this view, the role of communication is not to generate deictic structure, but to make use of it. Deixis is independently required as part of the grammatical system that anchors predication and yields propositional content (Martin & Hinzen 2014). Once linguistic representations are externalized into signals, however, the resulting underdetermination makes such anchoring indispensable for successful interpretation. Communication thus exploits deictic structure to reduce the uncertainty introduced by externalization. This perspective is thus compatible with the view that the faculty of language is primarily optimized for the conceptual-intentional interface, with externalization playing a secondary role (Berwick & Chomsky 2016).

To understand why such anchoring must be grammatically encoded, we must consider the consequences of the type of representations over which grammar operates. As discussed in Section 2, grammar operates on intensional objects obtained through a process of deindexation, whereby representations are decoupled from their immediate dependence on the world. The crucial point here is that this transformation can be understood as encoding under constrained signal resources. The original representation contains rich information about the world, including the identity of referents, their spatial and temporal location, and their relation to the speaker. By contrast, the linguistic signal is necessarily limited: it is linear, temporally unfolding, and subject to constraints of efficient coding. The mapping from world-indexed representations to linguistic signals therefore involves a reduction in informational resolution. In information-theoretic terms, this means that the encoding process cannot be lossless: the signal does not preserve all the distinctions present in the original representation, but collapses multiple possible states of affairs into a single form. Deindexation, understood in this way, is not merely an abstraction from the world, but a process that necessarily introduces information loss.

If deindexation involves compression, then it follows that the encoding process is subject to entropy. The mapping from underlying representations (messages) to signals is non-bijective: multiple distinct inputs can give rise to the same output. In Shannon's terms, this implies that the conditional entropy $H(X_D | X_S)$ is greater than zero. The consequence of this is irreversibility. Given a signal alone, it is in general impossible to reconstruct the original message uniquely. The signal underdetermines the intended interpretation: it is compatible with multiple possible referents, times, locations, and perspectives. This underdetermination is not a contingent imperfection of language, but a necessary consequence of efficient coding. Any system that compresses rich representations into a limited signal space will exhibit this property. This type of trade-off between signal economy and interpretive ambiguity has been independently identified in information-theoretic approaches to communication systems. In particular, Fortuny & Corominas-Murtra (2011, 2013) show that mappings between signals and meanings are subject to constraints that

balance efficiency and recoverability, yielding structured systems in which ambiguity is not eliminated but regulated. From this perspective, the underdetermination observed in linguistic signals is not incidental, but a necessary consequence of optimizing communication under constrained resources (see also Ferrer-i-Cancho & Solé 2003).

Given that the signal underdetermines the message, successful interpretation requires more than decoding the signal itself. The hearer must supplement the signal with additional information in order to reconstruct a determinate interpretation. This is the recovery problem. Formally, if the encoding process introduces a non-zero amount of uncertainty, then recovering the original message requires information sufficient to reduce that uncertainty to a workable level. In information-theoretic terms, the system must provide access to the parameters that allow the hearer to select the intended interpretation from the set of possibilities compatible with the signal. Importantly, this recovery information cannot be arbitrary. It must target precisely the dimensions along which information was lost in the encoding process. The question, then, is: what kind of information is lost when world-indexed representations are compressed into linguistic signals? The crucial observation is that the information lost in deindexation concerns the anchoring of representations to the world. When a richly specified representation is transformed into a linguistic signal, what is systematically abstracted away is not conceptual content per se, but the coordinates that situate that content relative to a speaker and a context. These coordinates include (1):

- (1) a. The identity of the agent (who is speaking or being referred to);
- b. the temporal location of the event (when it occurs);
- c. the spatial location (where it occurs), and
- d. the perspective from which the situation is construed.

This can be seen from even the simplest examples. A sentence like *She left early* does not, by itself, determine who the referent of *she* is, what temporal interval counts as *early*, or relative to which temporal or perspectival frame the event is being evaluated. These parameters are not encoded in the lexical content alone, and yet they are essential for fixing a determinate interpretation. What encoding under constrained signal resources removes, then, is the link between descriptive content and its anchoring in a particular context of utterance.

If the information lost concerns anchoring, then the recovery mechanism must provide anchoring information. That is, it must specify how the content of the signal is to be situated relative to a speaker, a time, and a location. Crucially, this kind of information cannot be derived from the signal alone. By definition, it is what the signal fails to encode fully. Nor can it be supplied by arbitrary inferential processes, since the system must guarantee that interpretation is possible in real time and under minimal assumptions. The only available source for such information is the context of utterance itself, understood as structured by parameters such as speaker, addressee, time, and place. These parameters are precisely those traditionally associated with deixis. Since what is lost in the encoding process is specifically the anchoring of content to a speaker-relative context, any recovery mechanism must reintroduce precisely these parameters; no alternative type of information would suffice

to restore reference. These parameters define the minimal form of deictic anchoring required for interpretation, which is further elaborated in grammar into more complex forms of perspectival and propositional organization.⁴

We now reach the crucial step. If recovery requires access to contextual parameters, and if interpretation is systematically guided by grammar, then these parameters must be encoded in the grammatical system itself. This is exactly what we observe. Natural languages obligatorily encode features such as person, tense, and, in many cases, spatial orientation. These features do not contribute descriptive content in the usual sense; rather, they specify how the content of an expression is anchored relative to a context of utterance. Deictic features are therefore not merely optional lexical items, but structural components of grammar that implement the recovery mechanism required by compression. They provide the parameters needed to reconstruct a contextually anchored interpretation from an underspecified signal. We can thus identify deixis with the grammatical encoding of recovery information.

This structural role of deixis is reflected in its distribution across grammatical domains. In the nominal domain, determiners and related functional elements contribute to referential anchoring. In the verbal domain, tense and aspect locate events relative to a temporal frame. In the clausal domain, complementisers and related structures contribute to the anchoring of propositions relative to a perspective or evaluation context. These parallel systems can be understood as domain-specific implementations of a single underlying requirement: the encoding of recovery parameters that allow intensional content to be reconnected to the world.

To sum up, we have shown that, given (i) the necessity of deindexation for grammatical computation, and (ii) the inevitability of information loss under compression, linguistic systems face a recovery problem: the signal underdetermines the intended message. Solving this problem requires encoding information about how the content is anchored to a context. We have argued that this information is precisely what is expressed by deictic features, and that grammar provides a systematic means of encoding it. This dual role is not accidental: the same structural property that enables predication to yield propositional content also ensures that such content can be recovered under the conditions imposed by externalization. The conclusion is therefore not merely that deixis is important for language, but that it is a

⁴ To reinforce this idea, these parameters define the minimal form of deictic anchoring required to restore the link between linguistic content and a context of use, namely anchoring to speaker, time, and location. In the framework adopted here, this corresponds to what may be termed *immediate deixis* (Bühler's 1932 *origo*). However, such anchoring is not sufficient for full propositional interpretation. As argued by Hinzen & Sheehan (2015), grammatical systems further elaborate this basic anchoring into more complex forms of perspectival organisation, involving the structuring of reference across domains (e.g. nominal, verbal, and clausal) and the integration of subjectivity and point of view. This extended system, that they call *extended deixis*, underpins the formation of fully propositional representations. The argument developed in the main text is therefore concerned with deriving the necessity of minimal deictic anchoring from general constraints on encoding and recoverability; the richer organisation of deixis in grammar can be seen as a systematic elaboration of this minimal requirement.

necessary consequence of the computational architecture of grammar under general constraints on efficient encoding and transmission.

5. Consequences and implications

The analysis developed in the previous sections yields a number of consequences for how we understand linguistic meaning, the role of context, and the architecture of grammar. These consequences follow from the interaction between deindexation, information loss under constrained encoding, and the requirement that linguistic representations remain recoverable once externalized. Crucially, they also reflect the dual role of deixis: it is required both for the constitution of propositional meaning within the cognitive system and for its recoverability under externalization. In this sense, the proposal contributes to ongoing debates on the relation between grammatical structure, semantic interpretation, and general constraints on efficient representation.

A first consequence concerns the status of context-dependence in language. On the account proposed here, context-dependence is not a marginal property restricted to a class of indexical expressions, but a necessary feature of linguistic interpretation as such (Travis 1997; Recanati 2004, 2010). Since the encoding of linguistic signals does not preserve all aspects of the original representation, the signal alone cannot determine a unique interpretation. Recovering the intended meaning requires access to contextual parameters that anchor the expression to a specific situation of use. This entails that all linguistic expressions are, in principle, dependent on context for their interpretation. The familiar cases of indexicals and demonstratives (Kaplan 1989, Perry 1993) are therefore not exceptional, but simply instances where this dependence is overtly encoded. By contrast, expressions traditionally treated as context-independent, such as proper names or descriptive phrases, are only apparently so: their interpretation also relies on contextual anchoring, even if this anchoring is not lexically marked. In this sense, the distinction between context-sensitive and context-insensitive expressions collapses. What varies is not whether context is required, but how explicitly the relevant parameters are encoded. From this perspective, the pervasive underdetermination of linguistic meaning is not a defect to be repaired by pragmatic enrichment, but a direct consequence of the system's design.

A second consequence concerns the status of ambiguity. If encoding under constrained signal resources introduces irreversibility, then ambiguity is an unavoidable feature of the mapping between representations and signals. A given signal will, in general, be compatible with multiple possible interpretations, and disambiguation will require additional information beyond the signal itself. On this view, ambiguity is not a contingent imperfection of language, but a structural property that follows from general constraints on efficient encoding. Any system that seeks to minimize signal complexity while preserving expressive power will exhibit this kind of underdetermination. The role of context is therefore not to repair an otherwise defective system, but to complement a system that is optimally organized given competing pressures. In this respect, the present proposal aligns with information-theoretic approaches that derive ambiguity from least-effort principles. In particular,

accounts such as Fortuny & Corominas-Murtra (2013) show that ambiguity arises as a consequence of compression and the resulting irreversibility in the mapping between representations and signals. The present proposal extends this line of reasoning by locating ambiguity within a broader system of reference: on this view, ambiguity is not only a consequence of compression, but reflects a more general condition of underdetermination that requires the grammatical encoding of recovery parameters. At the same time, the proposal departs from purely communicative accounts (Tomasello 2008; Kirby et al 2008) in locating the solution within the grammatical system itself. Rather than treating context as an external resource that resolves ambiguity post hoc, grammar encodes the parameters required for disambiguation in the form of deictic features. In this sense, communication does not introduce the resources needed to resolve ambiguity, but exploits a system that is independently required for the constitution of propositional meaning.

A third consequence concerns the overall design of grammar. The interaction between efficient encoding and recoverability suggests that grammar must satisfy two distinct but related constraints: on the one hand, the need to operate over abstract, recombinable representations, and on the other, the need to ensure that these representations can be anchored to the world once externalized. The resulting system can be understood as optimized for the conceptual-intentional interface, in line with Berwick & Chomsky (2016), while incorporating structural provisions that satisfy the constraints imposed by externalization. In this sense, externalization is not the source of grammatical structure, but it is not inert either: once present, it imposes non-trivial requirements on how reference must be encoded. Deictic encoding is precisely where these two domains intersect. It is the same structural property that enables predication to yield propositional content that also ensures that such content can be recovered under conditions of underspecified transmission.

The proposal also yields a number of empirical predictions. First, it predicts that no natural language is expected to lack mechanisms for encoding anchoring parameters such as person, time, and location. While languages may vary in how these parameters are expressed, whether through morphology, syntax, or discourse structure, their presence should be universal, since they are required for recoverability. This expectation is consistent with typological work showing that all languages encode distinctions related to person and temporal reference, even when these are not morphologically marked in a uniform way (e.g. Comrie 1985; Bhat 2004). Second, it predicts that even expressions that appear to be context-independent will exhibit context-dependence under closer scrutiny. Proper names, for instance, do not fix reference independently of context, but rely on contextual information for their interpretation, as widely discussed in the philosophical literature on reference (e.g. Kripke 1980). Third, it predicts that ambiguity will be pervasive and unavoidable, rather than confined to particular constructions. Attempts to eliminate ambiguity entirely would require an expansion of the signal space that runs counter to general constraints on efficient encoding. Finally, it predicts that disruptions to the encoding or availability of deictic parameters should lead to systematic difficulties in interpretation. If deixis provides the mechanism for anchoring linguistic representations, then impairments affecting this system should manifest as deficits in the ability to relate linguistic expressions

to a context of use, a pattern that is consistent with observations from clinical linguistics and psychopathology (e.g. Hinzen & Rosselló 2015).

Taken together, these consequences support a view of grammar in which reference is not an external relation established after the fact, but an integral part of the system's architecture. Grammar does not merely combine concepts into structured representations; it provides the means by which those representations are anchored to the world to create propositions. Deixis is therefore not a peripheral or optional component of language, but a structural requirement that follows from the interaction between the internal organization of grammar and the constraints imposed by its externalization. The pervasive context-dependence of linguistic meaning thus reflects a fundamental property of the system itself.

6. Conclusion

In this paper, we have proposed a principled account of deixis as a structural property of grammar that is independently required for propositional meaning, and whose role becomes indispensable under the constraints imposed by externalization. Starting from the observation that grammar operates over intensional representations, we argued that the transformation of world-indexed representations into linguistic signals requires encoding under constrained signal resources. This transformation does not preserve all aspects of the original representation, yielding irreversibility in the mapping between representations and signals. As a result, linguistic signals systematically underdetermine the intended interpretation, giving rise to a recovery problem: successful comprehension requires access to information that restores the link between linguistic content and the world.

We have argued that this information concerns the anchoring of representations to a context of use, and that grammar provides a systematic means of encoding it through deictic features. On this view, deixis is not limited to a class of indexical expressions, but constitutes a structural property of grammar itself. The encoding of parameters such as person, time, and location reflects the system's need to make available the anchoring information required for interpretation under conditions of underspecified transmission. More generally, the pervasive context-dependence of linguistic meaning follows from general constraints on representation and externalization, rather than from language-specific adaptations.

Crucially, this account reveals that deixis plays a dual role in the architecture of language. On the one hand, it is required for the constitution of propositional meaning, providing the anchoring that allows predication to yield truth-evaluable content within the cognitive system. On the other hand, it ensures that such content remains recoverable once linguistic representations are externalized into signals that underdetermine their interpretation. This dual role is not accidental. The same structural property that enables predication to yield propositional content also ensures that such content can be anchored and interpreted under the conditions imposed by externalization. In this sense, the role of deixis in anchoring predication is independent of its role in communication, and reflects a fundamental property of the cognitive organization of grammar, even though communication exploits this property to ensure successful interpretation.

This perspective is compatible with the view that the faculty of language is primarily optimized for the conceptual-intentional interface, with externalization playing a secondary role. At the same time, it suggests that externalization, once present, imposes non-trivial constraints on the form of grammatical systems. In particular, the need to recover reference from an underspecified signal forces the grammatical system to encode anchoring information explicitly. Communication, on this view, does not give rise to deictic structure, but makes use of a system that is independently required for the constitution of propositional meaning.

More generally, the proposal contributes to the broader biolinguistic goal of explaining the properties of language in terms of general constraints on computation and representation, rather than language-specific stipulations. By deriving the systematic role of deixis from independently motivated constraints on encoding and recoverability, we have argued that a central feature of linguistic structure can be understood as a necessary consequence of the system's design. Future work should develop the formal aspects of this proposal in greater detail, in particular by making explicit the relationship between grammatical structure and information-theoretic notions such as entropy and recoverability. It should also explore the empirical consequences of the proposal across languages and domains, including typological variation in the encoding of deictic features and the effects of disruptions to anchoring mechanisms in clinical populations. In this way, deixis is not merely a feature of linguistic expressions, but a necessary property of any system that must both constitute and externalize propositional meaning under general constraints on representation and transmission.

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